

Digitally Tuned AM Radio

Capacitor Switching with Arduino and Analog Switch ICs

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1. Introduction

This report documents the design and experimentation of a digitally tuned AM radio receiver front-end. A rotary encoder allows the user to step through capacitor combinations, selecting the total capacitance presented to the LC tank and thus tuning the receiver to different carrier frequencies.

Due to time constraints, the physical PCB was designed in Altium Designer but was not received before the project deadline. Validation was performed on a breadboard prototype using a Digilent Analog Discovery 3 (AD3) as both signal source and measurement instrument. The AD3 Wavegen was used to inject a synthetic AM-modulated carrier directly into the tank node, substituting a real broadcast signal. The AD3's Network Analyzer tool was used to perform frequency sweeps across the tank circuit, directly measuring the resonant frequency at each capacitor setting.

The system successfully demonstrates digitally controlled resonant frequency selection across a measured carrier range of approximately 275 kHz to 700 kHz and recovers AM-modulated audio content up to approximately 10 kHz bandwidth. Challenges encountered during the project included measuring the inductor self-resonant frequency, stray capacitance effects on resonant frequency, switch parasitic effects, and op-amp biasing.

2. System Architecture

The receiver is organized as a cascade of functional blocks, each responsible for one stage of signal processing. The signal flow from antenna to speaker is as follows:

Antenna (Loopstick) > LC Tank (Tuned) > Diode Detector > MCP6002 Op-Amp > LM386 Amplifier > Speaker

The LC tank is formed by the loopstick inductor in parallel with the digitally selected capacitor combination. The diode detector (1N5711 Schottky) extracts the audio envelope from the AM carrier. The MCP6002 op-amp buffers and amplifies the detected signal while biasing it around a mid-rail reference voltage ($V_{REF} \approx 2.5$ V) for single-supply operation. The LM386 provides the final audio power amplification to drive a small speaker. The Arduino Uno serves as the digital controller, reading a rotary encoder and translating the encoded position into a 7-bit binary value that selects which capacitors in the bank are active. Two SN74LV4066 quad analog switch ICs implement the physical switching of the seven capacitors.

3. Theory of Operation

3.1 LC Tank Resonance

An LC tank circuit resonates at a frequency determined by the inductance L and capacitance C according to:

$$f_0 = 1 / (2\pi\sqrt{LC})$$

At resonance, the impedance of the tank is maximized, so a signal at exactly the resonant frequency experiences the least attenuation through the tank. Signals at other frequencies are attenuated. This frequency selectivity is the basis of radio tuning.

The quality factor Q of the tank determines the sharpness of this selectivity. For a tank with series resistance R :

$$Q = \omega_0 L / R = 2\pi f_0 L / R$$

A higher Q means a narrower bandwidth and better rejection of adjacent channels. In this design, the on-resistance (R_{on}) of the 4066 analog switches adds series resistance to each capacitor branch, reducing Q . The SN74LV4066 has a typical R_{on} of approximately 100–200 Ω at 5 V, which is a known limitation of the switched-capacitor approach.

3.2 Capacitor Bank Tuning

Seven capacitors with values 2 pF, 4.7 pF, 8 pF, 16 pF, 39 pF, 82 pF, and 150 pF are individually switched into the tank via analog switches. When a switch is closed, that capacitor connects from the TANK node to ground, adding its capacitance in parallel to the tank. The total switchable capacitance is the sum of all enabled capacitors:

$$C_{total} = \sum C_i \text{ (for all enabled switches } i)$$

With 7 independent binary switches, there are $2^7 = 128$ possible combinations, spanning total capacitances from near 0 pF (all off) to approximately 301.7 pF (all on). The Arduino encodes the current selection as a 7-bit integer (0-127), where each bit corresponds to one capacitor. The rotary encoder increments or decrements this value, and the Arduino drives the switch control pins accordingly.

In practice, the resonant frequency is always lower than the ideal prediction from $f_0 = 1/(2\pi\sqrt{LC})$ because stray capacitance from the breadboard, switch off-state capacitance, and PCB parasitics add to the switched capacitance. This stray capacitance is characterized and quantified in Section 7.

3.3 AM Envelope Detection

Amplitude modulation encodes audio information as variations in the amplitude (envelope) of a high-frequency carrier. An AM signal can be written as:

$$s(t) = A_c[1 + m \cdot x(t)] \cdot \cos(2\pi f_c t)$$

where $x(t)$ is the audio message signal, m is the modulation index, f_c is the carrier frequency, and A_c is the carrier amplitude. The diode detector (envelope detector) recovers $x(t)$ by rectifying and low-pass filtering the received signal. The RC time constant of the detector must be chosen to follow the audio envelope but reject the carrier:

$$1/f_c \ll RC \ll 1/f_{au} \lambda_o$$

In this design, a 47 kΩ load resistor and 1 nF filter capacitor give a time constant of 47 μs, corresponding to a detector audio bandwidth of approximately $1/(2\pi \times 47\mu) \approx 3.4$ kHz. Measured results confirmed recoverable audio to approximately 10 kHz at lower carrier frequencies where the tank provides more voltage gain.

4. Hardware Implementation

4.1 Schematic Overview

The complete schematic is shown in Figure 1. The design is organized into functional regions: header pin connections for Arduino interfacing, the rotary encoder, VREF biasing, the inductor/speaker connector, two capacitor switching blocks, the op-amp stage, and the LM386 amplifier with detector network.

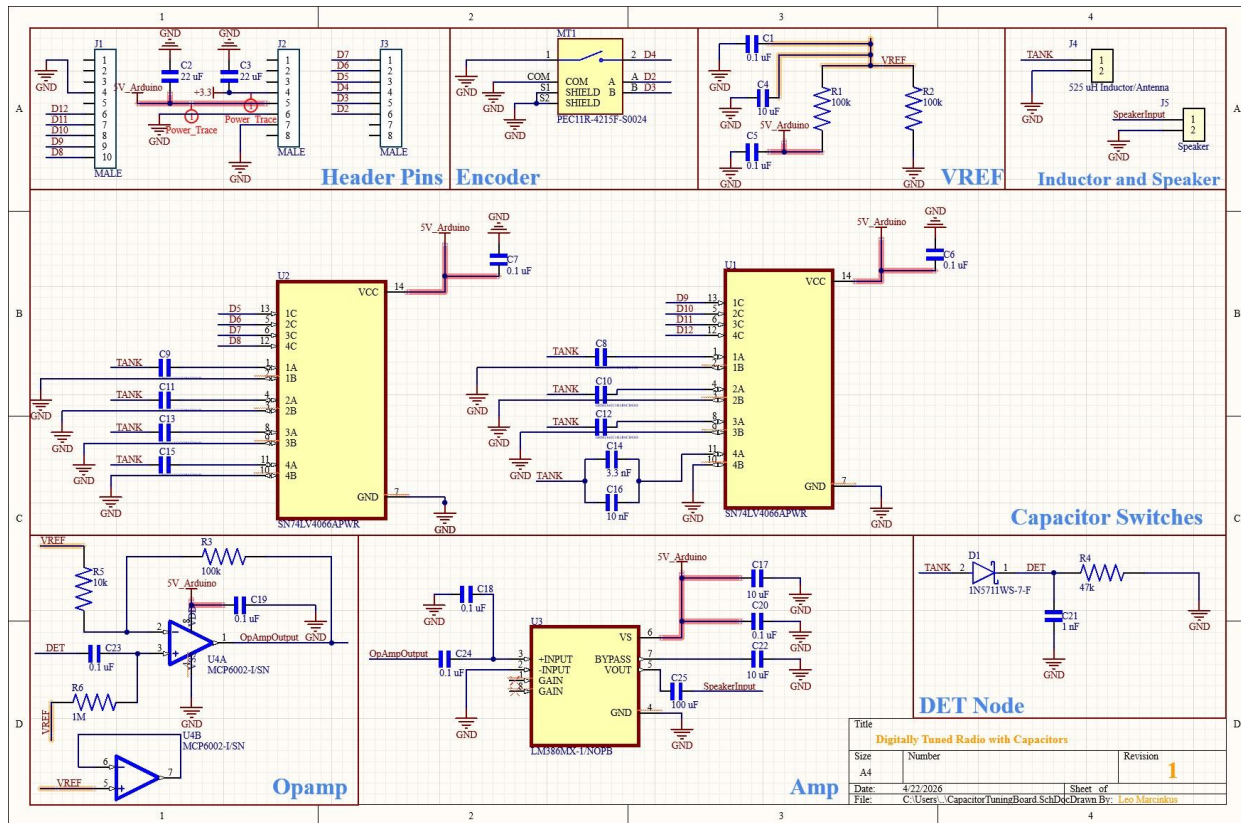


Figure 1: Complete circuit schematic designed in Altium Designer.

4.2 Component Selection

The key components and their selection rationale are summarized in Table 1.

Component	Part Number	Rationale
Analog Switches (x2)	SN74LV4066APWR	Quad SPST, 5 V compatible, $\sim 100\ \Omega$ R_{on} at 5 V
Detector Diode	1N5711WS-7-F	Schottky diode
Op-Amp	MCP6002-I/SN	Dual op-amp, rail-to-rail, single 5 V supply compatible
Audio Amplifier	LM386MX-1	Integrated audio power amp, drives $8\ \Omega$ speaker, adjustable gain
Inductor/Antenna	525 μ H Loopstick	Ferrite loopstick, serves as both inductor and AM antenna
Encoder	PEC11R-4215F-S0024	Rotary encoder with push-button switch

Table 1: Key component selection summary.

4.3 Analog Switch ICs (SN74LV4066)

The SN74LV4066 contains four independent SPST analog switches, each controlled by a digital logic input. Two ICs are used to accommodate all seven capacitors. Figure 2 shows the pin diagram. The A and B pins are the analog signal pins (bidirectional), and C is the digital control input. When C is high, the A-B path is closed; when C is low, the path is open.

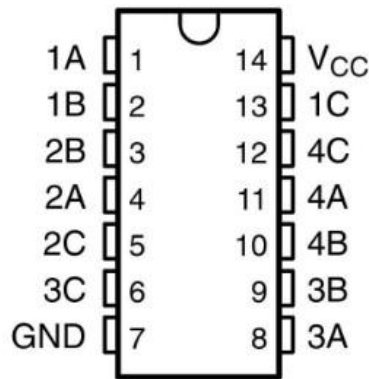


Figure 2: SN74LV4066 pin diagram.

Each capacitor is wired with one leg to the TANK node and the other leg to the A pin of its assigned switch. The B pin connects to ground. This places the capacitor in series with the switch, forming a switched shunt branch across the tank. When the switch closes, the capacitor is effectively connected from TANK to GND, adding its capacitance to the tank in parallel. When open, the capacitor is isolated. One channel on switch 2 is unused, its control pin is held low, and both analog pins are grounded to prevent any floating node from injecting noise.

4.4 Op-Amp Biasing (MCP6002)

The MCP6002 is a dual op-amp configured as a single-supply AC-coupled amplifier. A mid-rail reference voltage $V_{REF} \approx 2.5\text{ V}$ is generated by a resistor divider (two $100\text{ k}\Omega$ resistors from 5 V to GND) and decoupled with $10\text{ }\mu\text{F}$ and $0.1\text{ }\mu\text{F}$ capacitors. Figure 3 shows the MCP6002 pin diagram.

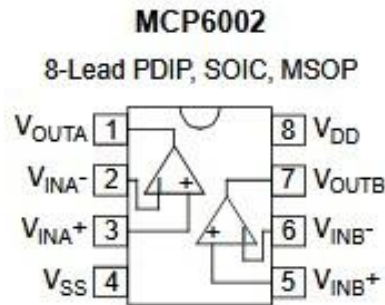


Figure 3: MCP6002 pin configuration.

Op-amp A is configured as a non-inverting amplifier with gain ≈ 11 . The DET node is AC-coupled to pin 3 through a $0.1\text{ }\mu\text{F}$ capacitor, with a $47\text{ k}\Omega$ resistor to V_{REF} establishing the DC bias. A $10\text{ k}\Omega$ resistor from pin 2 to V_{REF} and a $100\text{ k}\Omega$ feedback resistor complete the gain network. Op-amp B is configured as a unity-gain buffer with its non-inverting input tied to V_{REF} and pins 6 and 7 shorted, preventing oscillation from a floating input. A $0.1\text{ }\mu\text{F}$ decoupling capacitor is placed directly at the MCP6002 supply pin.

4.5 LM386 Audio Amplifier

The LM386 provides the final audio amplification stage. Figure 4 shows its pin diagram. The device operates from 5 V (pin 6) and drives the speaker through a $100\text{ }\mu\text{F}$ output coupling capacitor (pin 5). The inverting input (pin 2) is grounded and pins 1 and 8 (GAIN) are left open, setting default gain of 20. The supply is decoupled with $10\text{ }\mu\text{F}$ and $0.1\text{ }\mu\text{F}$ capacitors to GND, and a $10\text{ }\mu\text{F}$ capacitor on the bypass pin (pin 7) reduces internal noise.

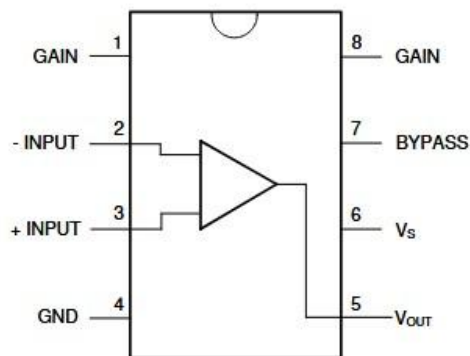


Figure 4: LM386 audio amplifier pin diagram.

5. Arduino Control Software

The Arduino firmware manages the rotary encoder and translates the encoder position into switch control signals. The core state is a 7-bit integer (0–127) where each bit corresponds to one capacitor. The encoder increases or decreases this value on each detent, and the firmware drives each switch control pin high or low according to the corresponding bit.

The mapping from bit position to physical Arduino pin and capacitor value is shown in Table 2.

Bit	Capacitor	Arduino Pin	Switch IC	Switch Channel
0	2 pF	D5	U2 (SW1)	1A/1B
1	4.7 pF	D7	U2 (SW1)	2A/2B
2	8 pF	D6	U2 (SW1)	3A/3B
3	16 pF	D8	U2 (SW1)	4A/4B
4	39 pF	D11	U1 (SW2)	1A/1B
5	82 pF	D10	U1 (SW2)	2A/2B
6	150 pF	D9	U1 (SW2)	4A/4B

Table 2: Capacitor bit mapping to Arduino pins and switch channels.

The encoder uses falling-edge detection on channel A, with channel B sampled at the moment of the falling edge to determine direction. A 5 ms delay is applied after each step to debounce the mechanical contacts. The value wraps at 0 and 127 for continuous rotation. On each change, the firmware prints active capacitors and computed total capacitance to serial at 115200 baud:

```
void applyCaps(int val) { float totalCap = 0.0; for (int i = 0; i < 7; i++) {
bool on = (val >> i) & 0x01; digitalWrite(caps[i], on ? HIGH : LOW); if (on)
totalCap += capValues[i]; } // Serial output: active caps + total pF }
```

Code Excerpt 1: Core capacitor switching function.

6. Inductor Characterization

Before attempting signal measurements, the loopstick inductor was characterized using the AD3 Network Analyzer. A 100 Ω series resistor was placed between W1 and the inductor, with the scope channel probing the junction node, forming a voltage divider whose frequency response reveals the inductor's characteristics.

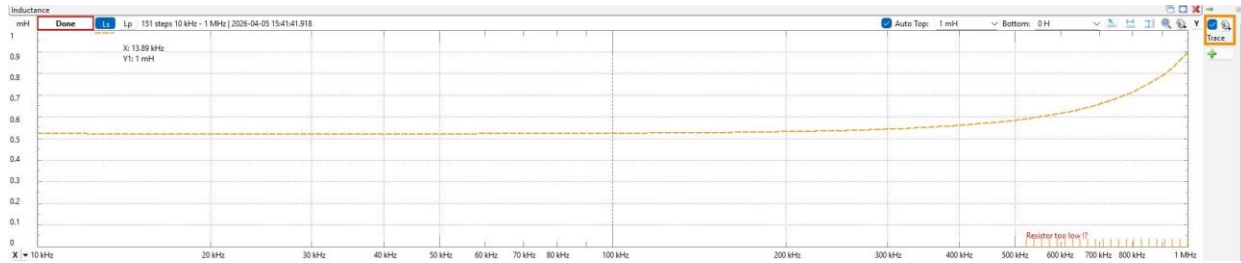


Figure 5: AD3 Impedance Analyzer inductance measurement showing approximately 525 μH at low frequencies rising toward the self-resonant frequency.

The Network Analyzer sweep revealed a self-resonant frequency (SRF) of approximately 1.925 MHz for the main winding, with measured inductance of approximately 525 μH at frequencies well below SRF. Above the SRF, the inductor appears capacitive rather than inductive, making it unsuitable as a tank element.

A critical finding was that circuit's stray capacitance lowers the apparent SRF when the inductor is installed in the circuit. This was confirmed by measuring the SRF in isolation (1.925 MHz) versus installed on the breadboard (as low as 914 kHz), a difference caused by approximately 15-20 pF of breadboard stray capacitance. This effect also shifts all tank resonant frequencies lower than the ideal calculations.

7. Experimental Results

7.1 Test Setup

All measurements were performed on the breadboard prototype. The AD3 Wavegen (W1) output was injected into the TANK node through a small series coupling capacitor (~ 8 pF) to weakly couple the source without significantly loading the tank. The AD3 oscilloscope simultaneously probed the TANK node (Channel 1) and the DET node (Channel 2).

Two types of measurements were performed:

- Time-domain: A synthetic AM-modulated carrier was injected and the oscilloscope captured TANK and DET waveforms, demonstrating envelope detection.
- Frequency-domain sweep: The AD3 Network Analyzer swept the tank from ~ 100 kHz to 1.5 MHz with no modulation directly measuring the resonant frequency peak at each capacitor combination. This allowed direct comparison of predicted versus measured resonant frequencies.

7.2 Resonant Frequency Sweep - Predicted vs. Measured

For each capacitor combination, the expected resonant frequency was first calculated using the ideal LC resonance formula with $L = 525 \mu\text{H}$ and the nominal switched capacitance only (no stray):

$$f_0 (\text{ideal}) = 1 / (2\pi\sqrt{L \times C})$$

This ideal prediction assumes a perfect inductor and capacitors only, with no parasitic effects. In practice, the resonant frequency is always lower due to stray capacitance. By measuring the tank with no capacitors switched in (0 pF case), the total stray capacitance of the breadboard circuit can be estimated. The measured resonant frequency of 560.6 kHz with all switches open implies a stray capacitance of approximately 154 pF:

$$C_{\text{Stray}} = 1 / ((2\pi \times 560600)^2 \times 525\mu\text{H}) \approx 154 \text{ pF}$$

A corrected expected frequency was then calculated for each combination using $C_{\text{total}} = C_{\text{switched}} + 154 \text{ pF}$. Table 3 compares all three values.

Capacitor Combination	Switched C (pF)	Ideal f_0 (kHz)	Expected w/ Stray (kHz)	Measured f_0 (kHz)
None (stray only)	0			560.6
39 + 8 + 2 pF	49	992	488	483.0
82 + 16 + 2 pF	100	695	436	421.5
150 pF	149.7	568	399	384.2
150 + 39 + 8 + 2	199	493	370	338.8
150 + 82 + 16 + 2	250	439	346	314.8
All caps	301.7	400	325	297.1

Table 3: Predicted vs. measured resonant frequency at each capacitor setting. Ideal prediction uses $L = 525 \mu\text{H}$ and switched C only. Expected with stray adds 154 pF estimated stray capacitance. Measured values are from AD3 Network Analyzer sweeps.

The measured values consistently fall below even the stray-corrected predictions. This remaining discrepancy is attributed to the actual capacitor values being higher than nominal (large capacitors such as 82 pF and 150 pF can have +10% tolerance), and to additional loading from the diode and op-amp input connected to the TANK node. The trend is consistent where increasing switched capacitance lowers the resonant frequency predictably. This confirms that the digital tuning scheme functions as designed.

7.3 Frequency Sweep Plots

Figures 6 through 12 show the AD3 Network Analyzer output for each capacitor combination. In each plot, the orange trace (TANK) shows the tank frequency response with a clear resonant peak, and the blue trace (DET) shows the detector node response. The red vertical

cursor marks the measured peak frequency. The upper panel shows magnitude (dB) and the lower panel shows phase.

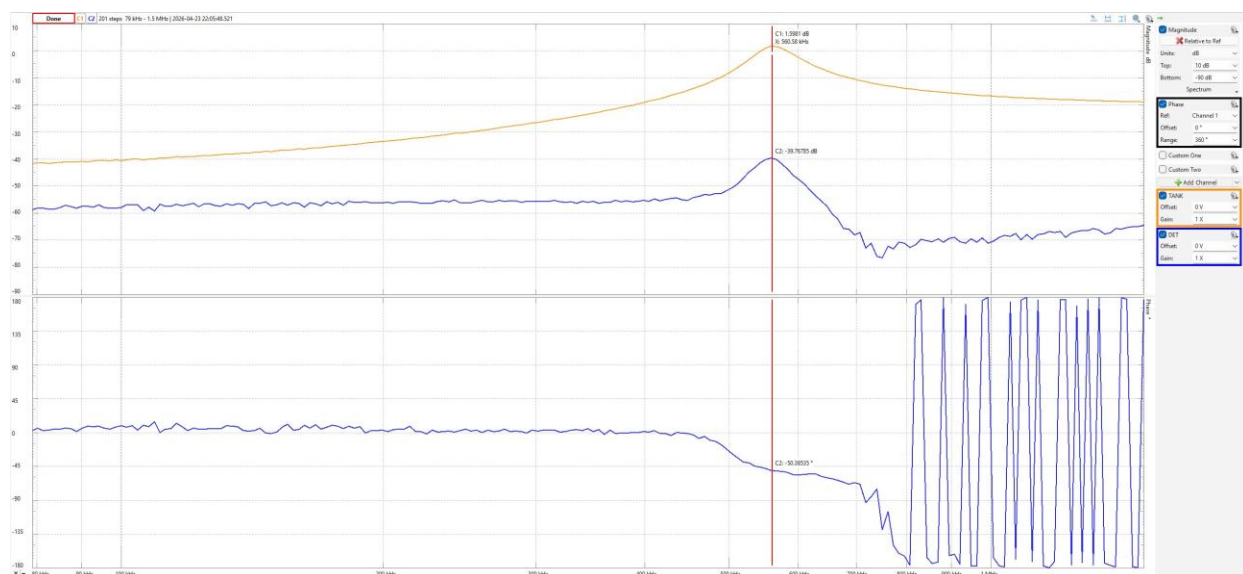


Figure 6: Frequency sweep with no capacitors switched in (0 pF). Peak at 560.6 kHz. This is entirely due to stray capacitance and inductor parasitics. Used to calibrate the stray capacitance estimate of ~ 154 pF.

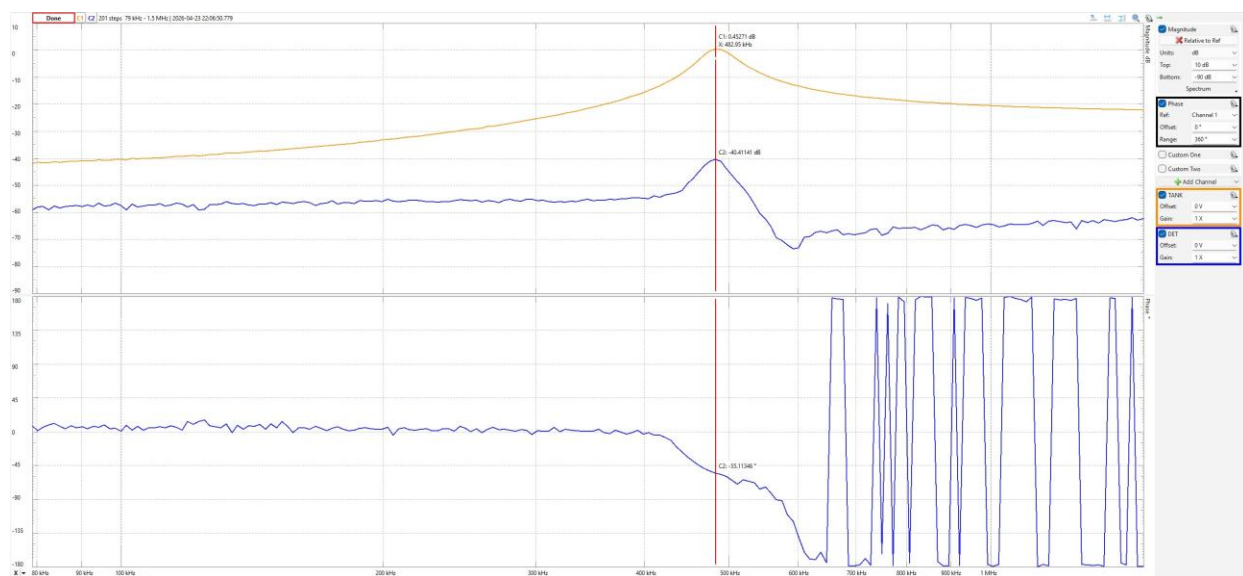


Figure 7: 49 pF switched in ($39 + 8 + 2$ pF). Measured resonant peak at 483.0 kHz. Ideal prediction: 992 kHz. Expected with stray: 488 kHz.

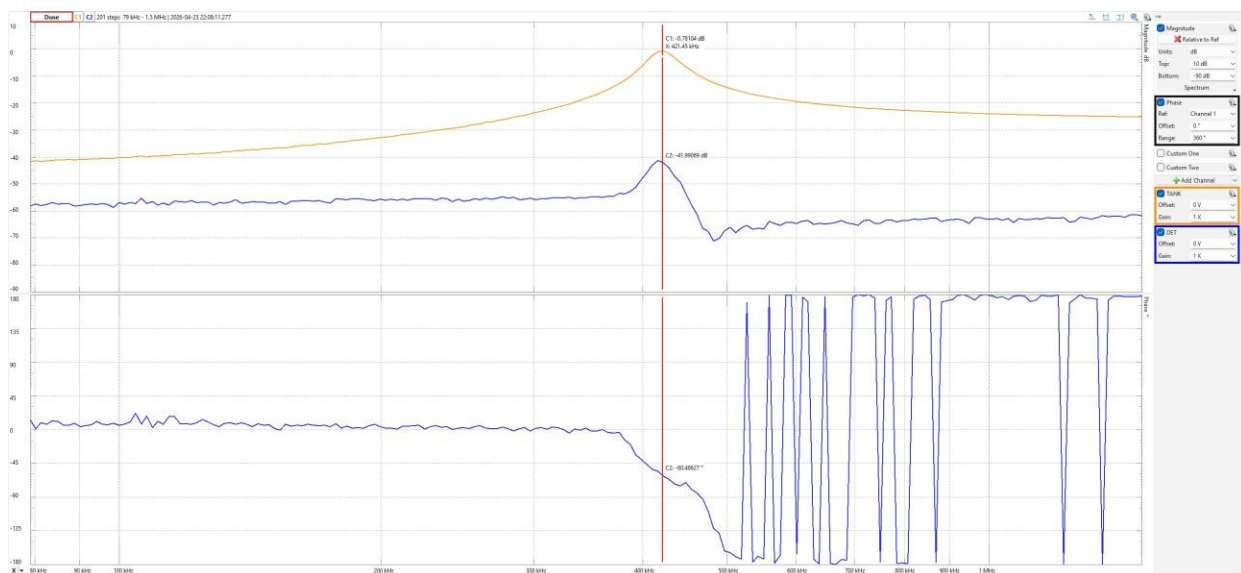


Figure 8: 100 pF switched in ($82 + 16 + 2$ pF). Measured resonant peak at 421.5 kHz. Ideal prediction: 695 kHz. Expected with stray: 436 kHz.

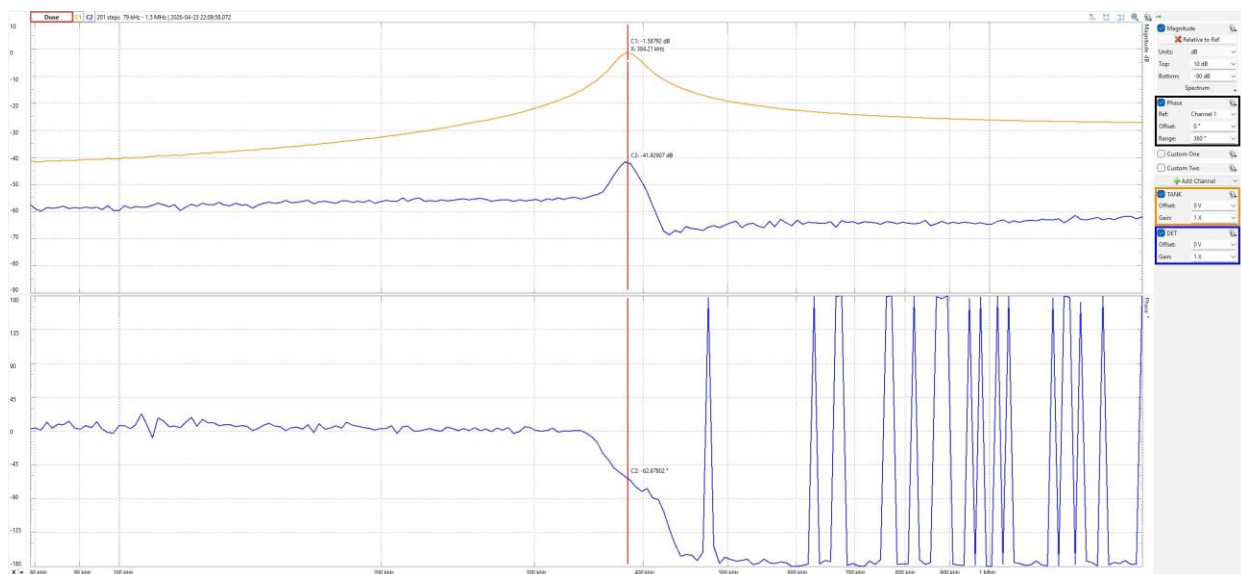


Figure 9: 149.7 pF switched in (150 pF). Measured resonant peak at 384.2 kHz. Ideal prediction: 568 kHz. Expected with stray: 399 kHz.

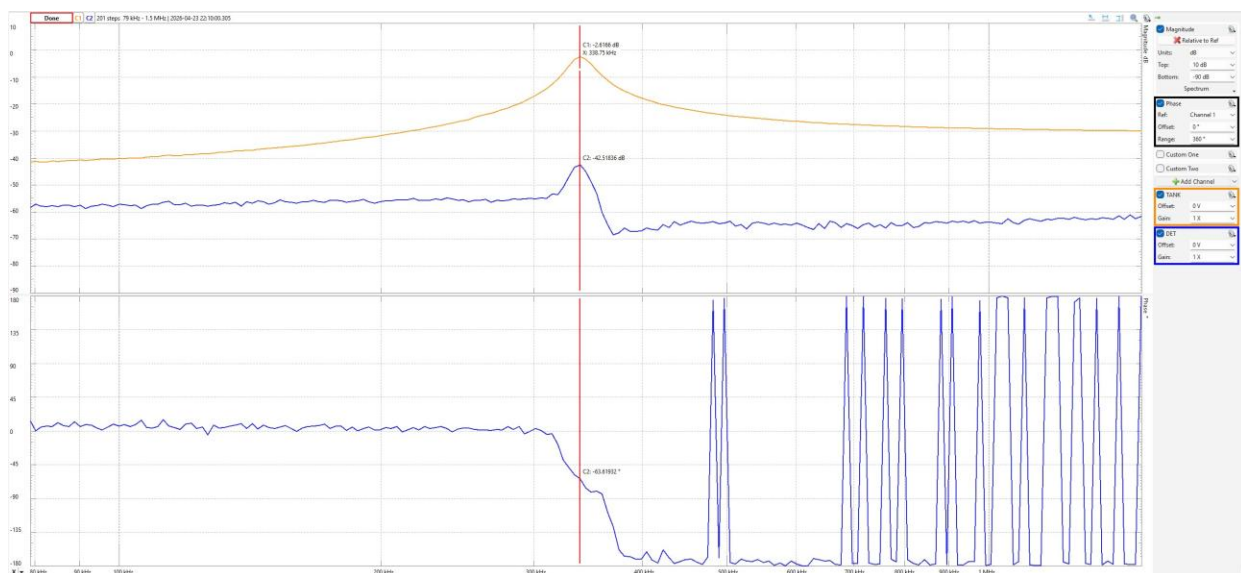


Figure 10: 199 pF switched in ($150 + 39 + 8 + 2$ pF). Measured resonant peak at 338.8 kHz. Ideal prediction: 493 kHz. Expected with stray: 370 kHz.

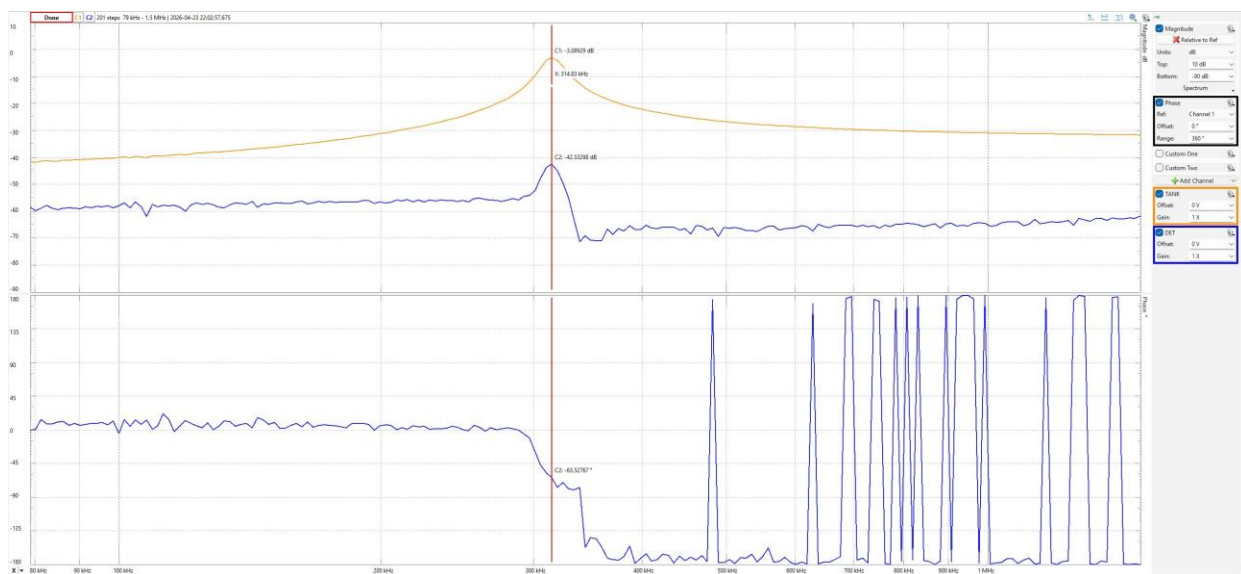


Figure 11: 250 pF switched in ($150 + 82 + 16 + 2$ pF). Measured resonant peak at 314.8 kHz. Ideal prediction: 439 kHz. Expected with stray: 346 kHz.

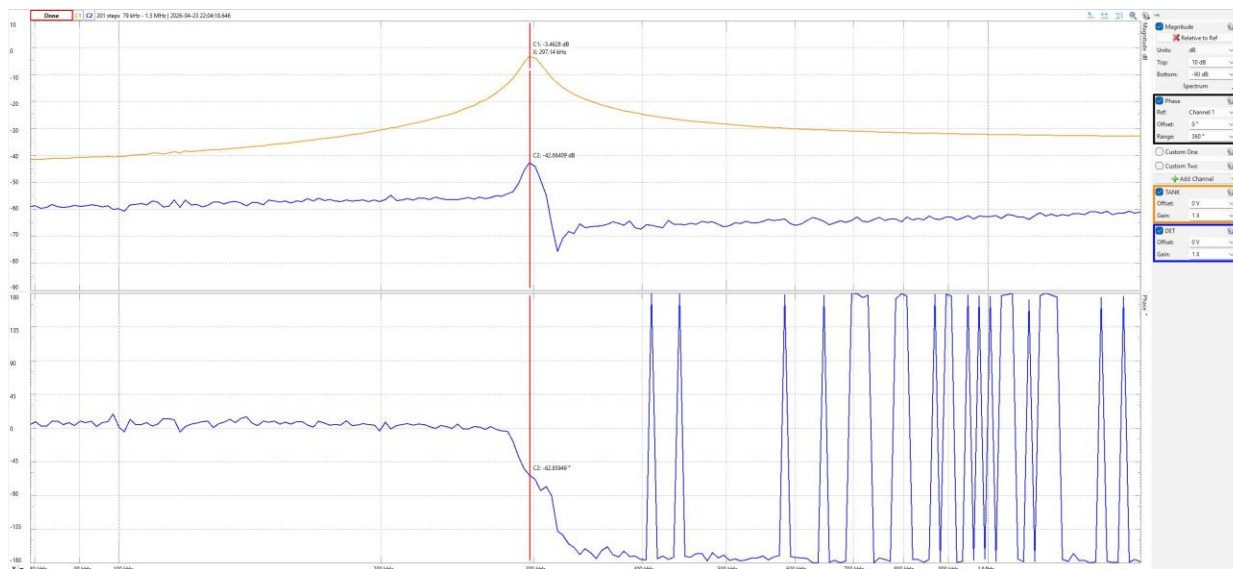


Figure 12: 301.7 pF switched in (all capacitors). Measured resonant peak at 297.1 kHz. Ideal prediction: 400 kHz. Expected with stray: 325 kHz. Lowest carrier frequency achieved.

7.4 AM Signal Injection and Time-Domain Detection

With the resonant frequencies characterized, an AM-modulated carrier was injected to demonstrate envelope detection. The AD3 Wavegen was configured as follows:

- Carrier type: Sine wave at 600 kHz
- Carrier amplitude: 300-800 mV peak
- Modulation type: AM, sine wave audio at 1 kHz
- Modulation depth: 50%

Figure 13 shows the AM modulation parameters configured in the AD3 Wavegen at 800 mV carrier amplitude.

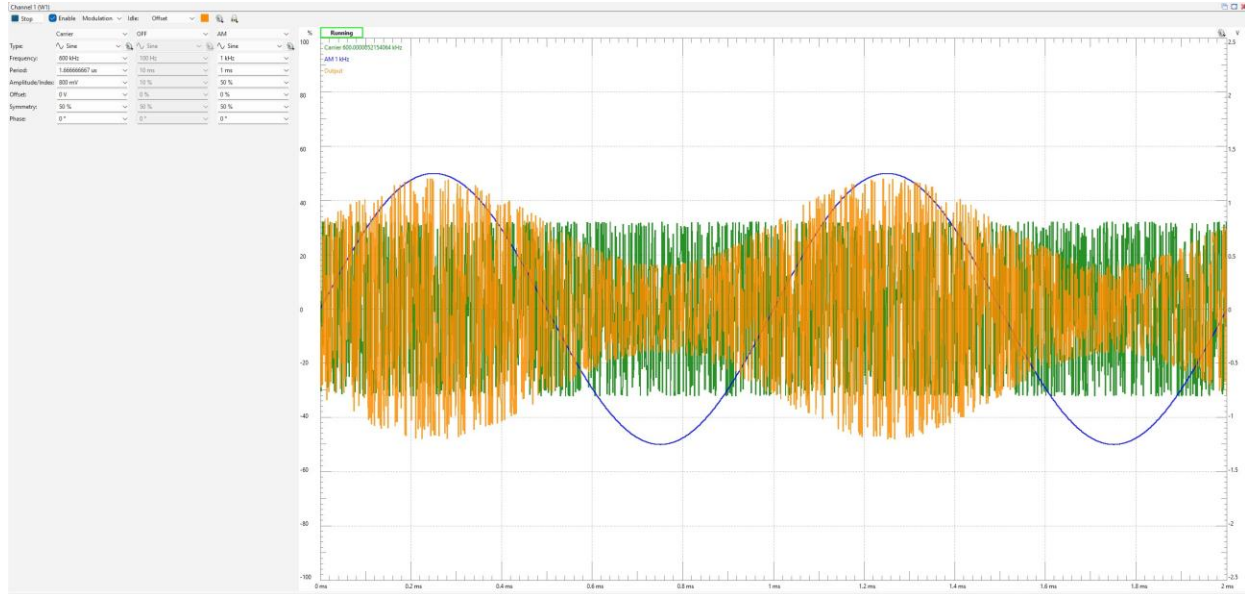


Figure 13: AD3 Wavegen AM modulation configuration. Carrier: 600 kHz, AM: 1 kHz at 50% depth, 800 mV amplitude.

Figures 14 and 15 show time-domain oscilloscope captures of the 26 pF configuration with a 600 kHz carrier. Figure 14 is zoomed out to show the 1 kHz audio envelope modulating the carrier amplitude, while Figure 15 zooms in to reveal the individual 600 kHz carrier cycles within the envelope.

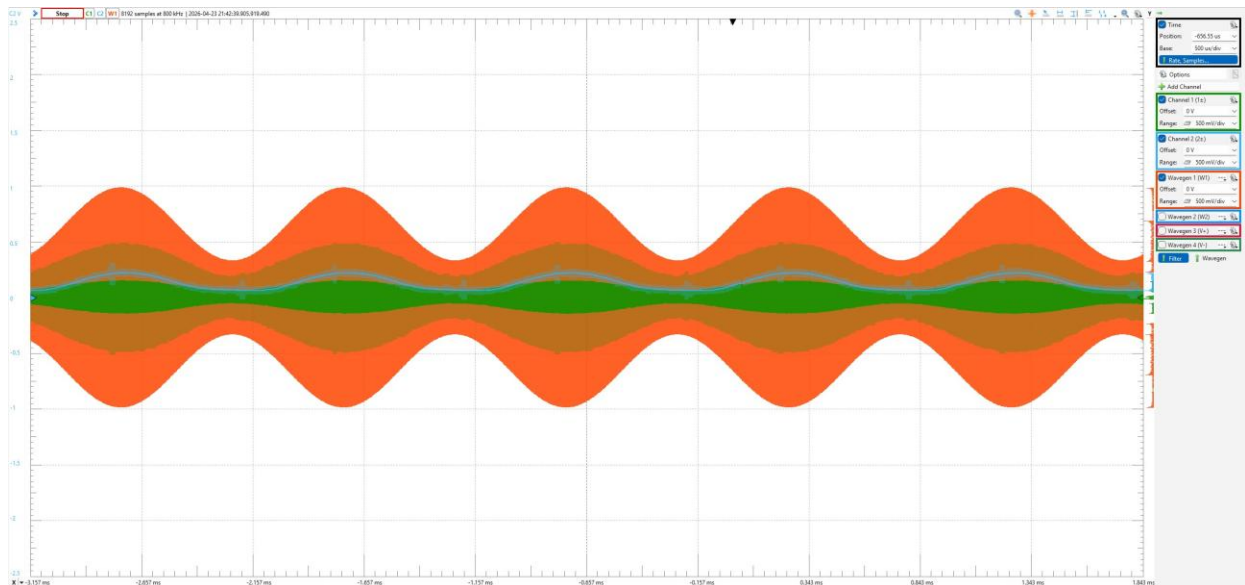


Figure 14: Time-domain capture, 26 pF at 600 kHz carrier with 1 kHz AM modulation, zoomed out. The orange/red traces show the AM envelope clearly at the 1 kHz audio rate. The blue trace near zero is the DET node output and the green trace is the TANK node.

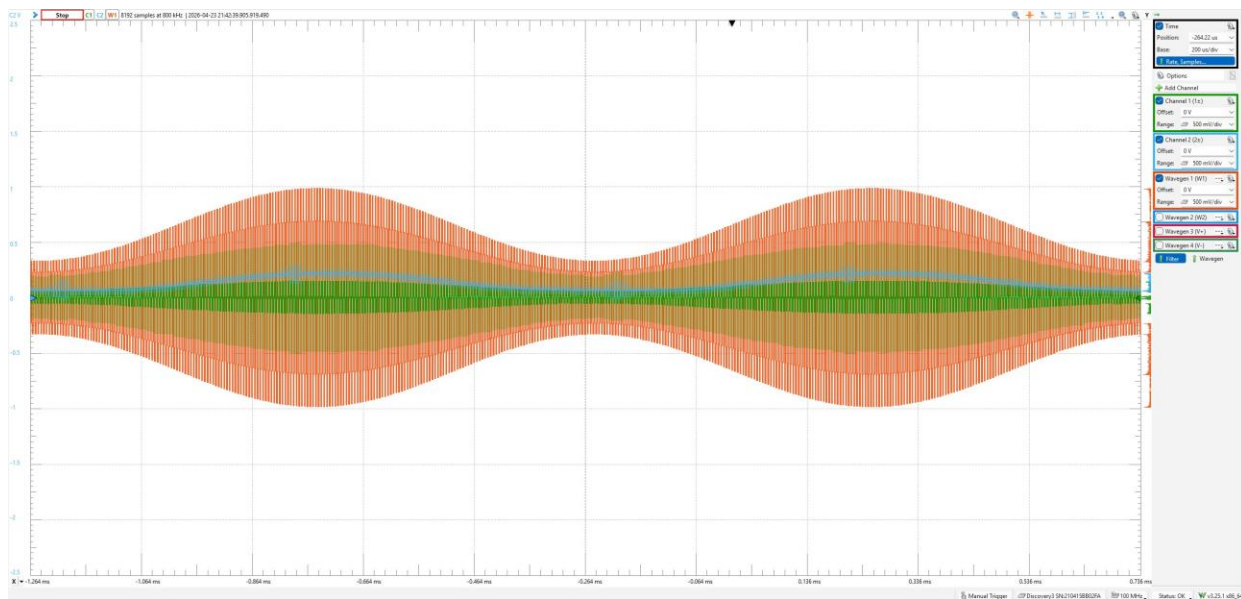


Figure 15: Same capture zoomed in. Individual 600 kHz carrier cycles are now visible within the amplitude-modulated envelope, confirming the carrier frequency and modulation are correct.

Figure 16 shows the best-performing configuration: 26 pF (16 + 8 + 2 pF) at 600 kHz carrier. Both the TANK node (orange, showing the AM-modulated carrier) and the DET node (blue, showing the recovered audio envelope) are clearly visible. This configuration produced the strongest detected signal of all tested settings.

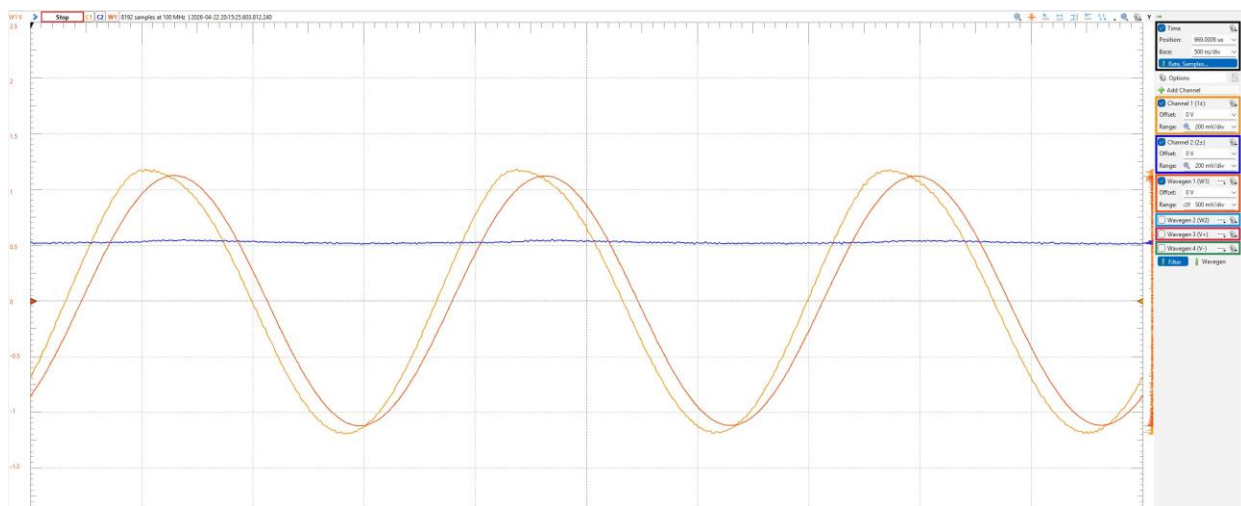


Figure 16: 26 pF (16 + 8 + 2 pF) at 600 kHz with 1 kHz AM modulation. TANK (orange) and detected envelope (blue) both clearly visible. 600 kHz injected waveform (red). Best-performing configuration.

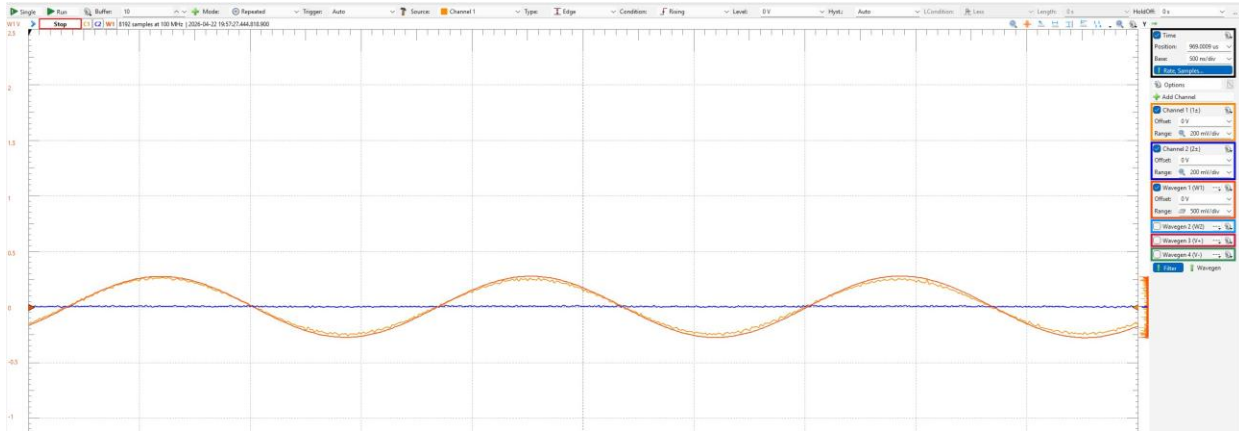


Figure 17: 28.7 pF (16 + 8 + 4.7 pF) at 600 kHz with 200 mV amplitude. Tank resonance visible but amplitude below the Schottky diode forward voltage threshold, resulting in no usable DET output.

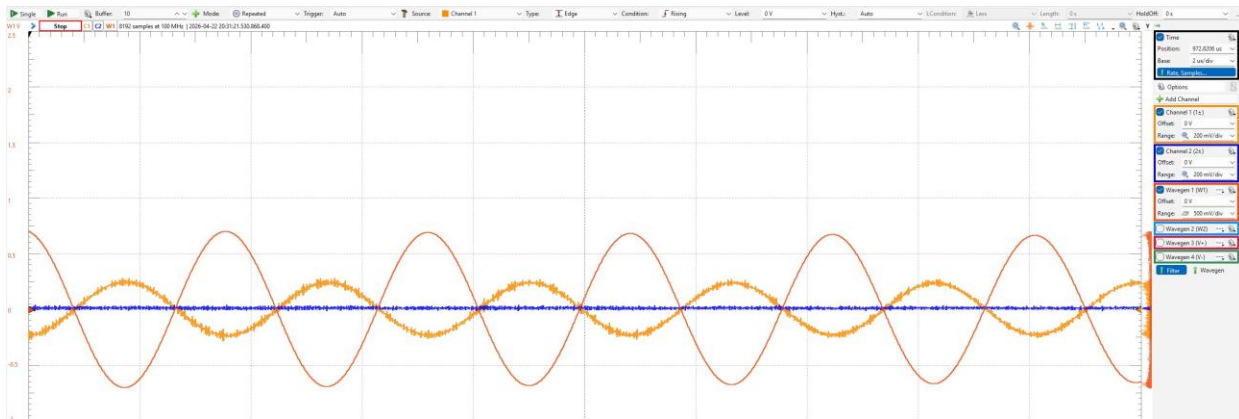


Figure 18: 301.7 pF (all capacitors) at 275 kHz carrier, demonstrating the lowest carrier frequency tested.

7.5 Audio Bandwidth Characterization

With the 26 pF combination at a 600 kHz carrier, the audio modulation frequency was swept to determine the maximum recoverable audio bandwidth. Results are summarized in Table 4.

Audio Modulation Frequency	DET Output	Result
1 kHz	Strong, clean envelope	Pass
5 kHz	Clean envelope	Pass
10 kHz	Detectable, reduced amplitude	Pass (marginal)
Above 10 kHz	Not detectable	Fail

Table 4: Audio modulation bandwidth test results at 600 kHz carrier, 26 pF tank.

The 10 kHz upper limit is consistent with the RC time constant of the detector network ($47\text{ k}\Omega \times 1\text{ nF} = 47\text{ }\mu\text{s}$, corresponding to a 3 dB frequency of approximately 3.4 kHz). The response extending to 10 kHz suggests the tank provides additional signal amplitude at lower audio frequencies. This bandwidth is adequate for AM broadcast audio fidelity, which typically occupies 5-8 kHz.

8. PCB Design

A two-layer PCB was designed in Altium Designer to implement the full receiver circuit. The PCB was designed and submitted for fabrication but was not received before the project deadline. Figures 19 and 20 show the PCB layout and 3D rendering.

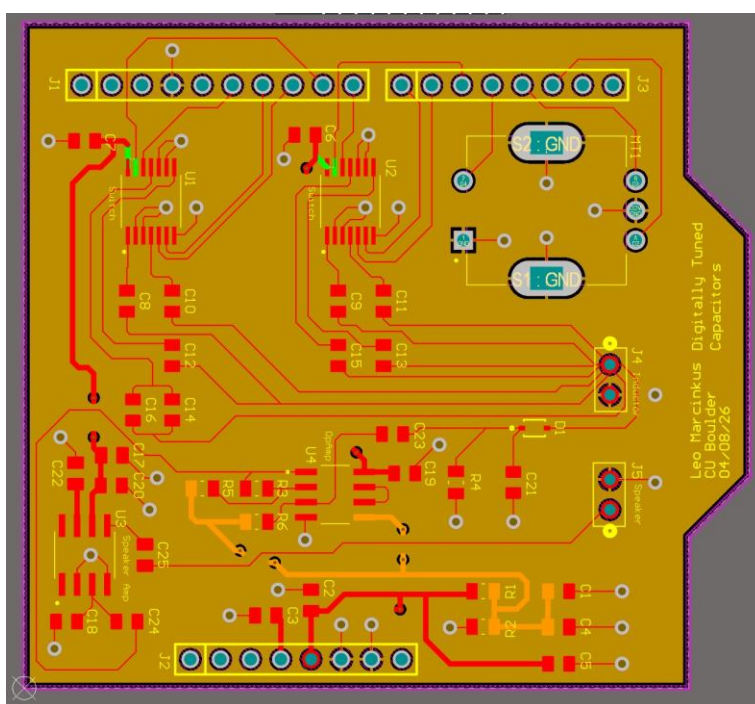


Figure 19: PCB layout in Altium Designer. The two switch ICs (U1, U2) and their capacitor arrays occupy the upper half. The analog signal chain occupies the lower section.

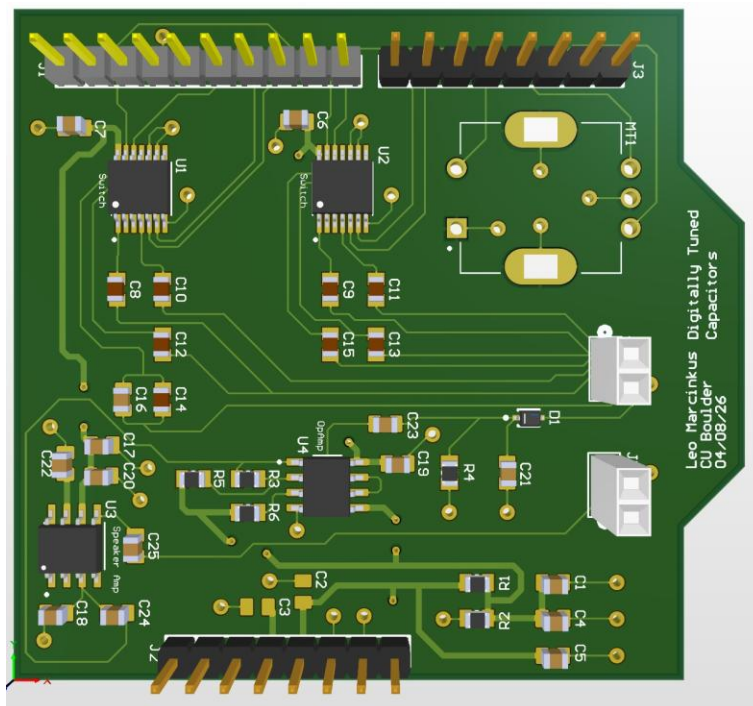


Figure 20: 3D rendering of the PCB design.

Key layout decisions include:

- Decoupling capacitors placed directly adjacent to IC supply pins to minimize supply noise coupling.
- The TANK node is a short, isolated trace connecting only the inductor connector, capacitor switch outputs, and diode anode, minimizing parasitic capacitance.
- Analog and digital signal traces are separated, with encoder and Arduino headers along the top edge and the analog chain in the lower half.
- VREF biasing network placed close to the MCP6002 to minimize noise on the reference node.
- The LM386 and its larger electrolytic capacitors are grouped in the lower-left section.

The PCB implementation is expected to reduce the stray capacitance (estimated at 154 pF on the breadboard) due to shorter trace lengths, absence of long interconnect wires, and a more controlled dielectric environment. This should shift the tuning range higher toward the AM broadcast band and improve tank Q.

9. Discussion

9.1 Stray Capacitance and Frequency Offset

The most significant finding from the frequency sweep measurements (Table 3) is that the measured resonant frequencies are consistently lower than both the ideal prediction and the stray-corrected prediction.

The 154 pF stray capacitance estimated from the 0 pF baseline measurement explains a large portion of the offset but not all of it. The remaining discrepancy at higher capacitance settings is most likely due to large capacitor tolerances (82 pF and 150 pF capacitors commonly have $\pm 10\%$ tolerance) and the loading effect of the diode and op-amp input on the TANK node. In a PCB implementation, the stray capacitance would be smaller and more predictable, making the theoretical predictions more accurate. The stray could also be characterized precisely and incorporated into the Arduino firmware to display a calibrated frequency readout.

9.2 Inductor Self-Resonant Frequency

The loopstick inductor's self-resonant frequency (SRF) was measured at approximately 914 kHz when installed on the breadboard, and 1.925 MHz in isolation. Since the target AM broadcast band (535-1705 kHz) lies above the breadboard-loaded SRF, the inductor could not be used above approximately 700 kHz on the breadboard prototype. On the PCB with reduced stray capacitance, the effective SRF should be significantly higher, potentially enabling operation in the AM broadcast band.

9.3 Q Factor Degradation from Switch R_{on}

The R_{on} of each closed switch adds series resistance to its capacitor branch. For $R_{on} \approx 150 \Omega$ and tank resonance at 400 kHz with 525 μH inductance:

$$Q = 2\pi \times 400 \times 10^3 \times 525 \times 10^{-6} / 150 \approx 8.8$$

This is considerably lower than the Q of 50-100 achievable with a mechanical variable capacitor. The frequency sweep plots confirm this: the resonant peaks in Figures 6-12 are broad, indicating low Q. This broad response means the tank accepts a wide range of frequencies, reducing selectivity. The effect is visible in the plots where the -3 dB bandwidth spans tens of kHz around each peak.

9.4 Single-Supply Op-Amp Biasing

Operating the MCP6002 from a single 5 V supply required careful DC biasing. The VREF divider (two 100 k Ω resistors) provides approximately 2.5 V, allowing the op-amp output to swing symmetrically above and below mid-supply. The 47 k Ω bias resistor on the non-inverting input, combined with the 10 k Ω feedback resistor to VREF on the inverting input, establishes the correct DC operating point. Without this biasing, the AC-coupled input would have no defined DC level and the output would saturate.

10. Conclusions

This project successfully demonstrated the core principle of a digitally tuned AM receiver using switched capacitors. The following objectives were achieved:

- A 7-bit capacitor bank spanning 2 pF to 301.7 pF was implemented using two SN74LV4066 quad analog switch ICs controlled by an Arduino.
- The rotary encoder interface allows tuning through 128 distinct capacitance combinations via a 7-bit binary encoding scheme.
- LC tank resonant frequencies were directly measured via AD3 Network Analyzer frequency sweeps at seven capacitor settings, spanning 297 kHz to 561 kHz.
- Measured resonant frequencies were compared to ideal predictions from $f_0 = 1/(2\pi\sqrt{LC})$, with discrepancies explained by a characterized stray capacitance of approximately 154 pF.
- AM envelope detection was demonstrated with recovered audio bandwidth up to approximately 10 kHz.
- A complete PCB was designed in Altium Designer incorporating all circuit stages with proper decoupling and layout practices.

The digital tuning approach is validated as a functional alternative to mechanical variable capacitors, with the tradeoffs of reduced Q (due to switch R_{on}) and discrete tuning steps.

11. Future Work

Several improvements are planned or recommended for future iterations:

- Fabricate and assemble the designed PCB. Reduced stray capacitance should extend the tuning range into the AM broadcast band (535-1705 kHz) and improve tank Q.
- Add an RF preamplifier stage between the antenna and the tank to boost the received signal before detection, improving sensitivity for real broadcast reception.
- Characterize the PCB stray capacitance using the same 0 pF baseline sweep technique, and incorporate the offset into the Arduino firmware to display a calibrated frequency readout.
- Evaluate lower- C_{off} analog switches (such as the Vishay DG611A, $C_{off} \approx 2\text{-}3$ pF) to reduce the stray capacitance contribution from inactive switch channels.
- Potentially use logic MOSFETs instead of analog switches to switch between capacitors.